



Cybersecurity Awareness and Communication

The case of CVEHunter

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ABSTRACT

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The case of CVEhunter

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The purpose of this thesis is to look deeper into cybersecurity awareness. How cybersecurity awareness can be increased among business entities. What kind of cybersecurity threats exist and challenges small and medium sized enterprises.

The aim of research is to reveal strengths and weaknesses of CVEHunter using qualitative and quantitative methods. A cybersecurity company offering automated vulnerability scanning services.

The theoretical framework drew on existing research on cybersecurity awareness, digital communication, social media marketing, and digital trust. Empirical data were collected during an internship at CVEHunter. Including social media analytics, website performance data, outreach activities.

Findings confirm that cybersecurity communication is most effective when messages are clear, concise, and directly connected to real-world threats. Interactive content, particularly polls and reels generate higher engagement. However, the results also show that awareness alone is insufficient without trust-building communication and a user-friendly website.

The study recommends that structured, transparent, and user-centered communication strategies play a key role in strengthening cybersecurity awareness among SMEs.

Key words: cybersecurity awareness, digital communication, trust, social media, SMEs

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1 INTRODUCTION

1.1 Background and starting point

Since most of the information is accessible online, protecting data has become an important worry for both tech giants and SMEs. Therefore, security sits front-and-center in their plans. As cyber threats keep shifting and the digital world expands, companies are under pressure to lock down their networks, and they must tell users what they're doing about security. May it be security patches or authenticators. Large companies usually have their own security teams, but SMEs like a printing shop miss the know-how and the money to even spot a threat, so, they might stumble on awareness.

CVEHunter, automatically scans for bugs on websites, APIs, servers, and cloud setups. CVEHunter's mission is to make discovering and fixing security concerns, That is one of the reason working on speeding up the site and push its search ranking higher is a necessity. This thesis uses CVEHunter as a case study and tries to show how online safety messages can reach wider audience, which being the case, trust grows; moreover, businesses start being safer.

Even though the need for cybersecurity awareness keeps growing, many firms still fumble when they try to put security advice into plain words. Technical terms and threats usually downgrade the understanding of the message. Consequently, many just skip the advice on keeping their computers safe.

1.2 Research problem and questions

Despite the growing need for cybersecurity awareness, many organizations still struggle to communicate security-related messages in a clear, accessible, and engaging way. Technical jargon, fear-based messaging, and information overload often reduce the effectiveness of cybersecurity related communication. This

research-based thesis aims to clarify how communication can influence awareness and increase trust among daily users and businesses entities.

Research Questions:

1. How can cybersecurity awareness be effectively communicated to professional and business audiences?
2. What communication strategies and content types are most effective in raising cybersecurity engagement?
3. How can digital channels such as LinkedIn, Facebook, and Xing support the communication of cybersecurity issues?

1.3 Objectives of the thesis

The primary objective of this thesis is to observe methods of cybersecurity communication. The study's aim is to inspect CVEHunter's communication practices. Identify key factors and based on these factors, propose practical recommendations.

Observing empirical findings and combining them with theoretical knowledge, this study seeks to contribute to the field of cybersecurity communications.

1.4 Research methods and data collection

Qualitative approach was utilized, with the integration of both theoretical factors and empirical findings. Empirical findings were gathered from the internship period at CVEHunter

Qualitative data:

- Social media engagement data (LinkedIn, Facebook, Xing, Reddit).
- Content strategy results (posts, polls, outreach metrics).
- Website analytics and user experience findings (GMetrix).
- Keyword research and SEO analysis.

Data was collected through analysis, Facebook and LinkedIn provided analytics, and other internal documents from CVEHunter.

AI tools, to be exact, ChatGPT, and Grammarly, may be used for **linguistic assistance, content structure, and preliminary thematic categorization**, ensuring efficient and coherent data presentation while maintaining academic integrity.

Content analysis was used to evaluate communication quality. **Comparative analysis** served the purpose of CVEHunter's information distribution of cybersecurity awareness. **Performance analysis** was conducted to examine important factors related to the website, such as click rates and website bounce rate.

1.5 Structure of the thesis

The thesis consists of seven main chapters.

- **Chapter 1 (Introduction)** presents the background, research problem, objectives, and methodological overview.
- **Chapter 2 (Theoretical framework)** introduces key concepts of cybersecurity awareness, digital communication, and trust-building in online contexts.
- **Chapter 3 (Research methodology)** describes the data collection and analysis methods in detail.
- **Chapter 4 (Empirical findings)** presents the results of the analysis, focusing on CVEHunter's communication strategies and performance.
- **Chapter 5 (Discussion)** evaluates the findings against theoretical insights and explores implications for practice.
- **Chapter 6 (Conclusions)** summarizes the results.
- **Chapter 7 (Recommendation)** provides actionable suggestions for CVEHunter and future studies.

2 THEORETICAL FRAMEWORK

2.1 Cybersecurity awareness and communication

Cybersecurity awareness is the understanding of digital threats and the knowledge for one to protect themselves. Including recognition, responding to threats, and solving the issue via digital practices. According to the European Union Agency for Cybersecurity, awareness is essential to modern cybersecurity strategies, as human error remains a leading cause of data breaches and security incidents. (European Union Agency for Cybersecurity., 2021)

The importance of cybersecurity awareness is crucial for small and medium-sized enterprises (SMEs). Many SMEs do not have resources to own cybersecurity departments and instead depend on external service providers or automated tools for protection. Based on these findings, it can be assumed that fostering awareness and understanding of common vulnerabilities is critical for maintaining operational security and customer trust. Awareness programs that incorporate practical examples, simulations, and clear communication have been shown to significantly enhance employees' capacity to identify and respond to cyber threats (Parsons et al. 2017)

2.1.1 Communication externally

Cybersecurity awareness also extends beyond internal training. It includes how organizations communicate about security externally (Alahmari et al, 2022). Companies must educate clients and the public on cybersecurity. In my findings I have found that the best practices not only reduce collective risk but also increase their brand public relation and reputation as responsible and trustworthy actors in the digital space.

According to Computer Science Review that technical skills in cybersecurity awareness relies heavily on consistent and comprehensible and clear communication (Chaudhary et al. 2023). Research shows that most individuals do not react to cybersecurity information unless it is communicated in a simple, meaningful, and context-relevant way (CISA, 2024). This means that awareness is not only a cognitive process, but also a communication challenge: users must first understand the message before they can act on it.

For SMEs, this is particularly important, as limited resources and a lack of dedicated security staff make effective communication one of the most impactful tools for risk reduction. According to these findings, it can be assumed that awareness only remains sustainable when security behaviours are continuously reinforced (ENISA, 2024).

From an SME perspective, cybersecurity awareness and communication must be understood in a simple manner. Where the goal is not just informing but teaching as well. So that user can recognise and react.

2.2 Digital trust and online safety communication

Effective communication means clear information distribution. To avoid miscommunication, especially in cybersecurity context, often balancing accuracy with simplicity helps users to understand technical information. ENISA's (2024) finding are in pair with this approach, examples, guide short videos will help the audience to decode companies' message

In the context of cybersecurity, communication splits into two forms:

- **Internal communication** building employee awareness, reporting incidents, and training.
- **External communication**, which seeks to inform customers, stakeholders, and the public about risks, company policies, or product security features.

The most important thing in common between internal and external communication is the challenge to reach experts and daily users. According to CISA (2024), relatable examples and clear steps have proved to lower risk mitigation when it comes to cybersecurity threats.

Trust is a crucial part of effective cybersecurity communication. With clear and timely communication following data breach, organizations can remain compliant.

2.3 Social media and B2B marketing in cybersecurity

In recent years, digital marketing platforms such as LinkedIn, Facebook, and Xing have emerged as valuable tools for raising cybersecurity awareness (Nurgül, 2024). Social media enables business entities to distribute and spread insights, deliver educational content.

Interactive formats, such as polls, short videos, and infographics, have demonstrated effectiveness in enhancing engagement and information retention (Edelman & LinkedIn, 2021).

In the case of CVEHunter, digital marketing fulfils both promotional and educational functions. Sharing vulnerability trends, security tips enable to establish credibility. Additionally, social media enables real-time feedback, allowing CVEHunter to evaluate which messages are most effective with its target audiences.

According to Bada et al. (2019), the success of awareness campaigns depends on several factors:

- **Content clarity:** The message must be clear, even for non-technical audiences.
- **Relevance:** Topics should point out current or emerging threats.
- **Consistency:** Continuous communication reinforces the organization's expertise and reliability.

- **Interactivity:** Engaging formats encourage users to participate and share information.

These parts together form the basis for effective cybersecurity awareness communication. These findings are supported by Critical Success Factors for Implementing Cybersecurity Awareness Programs in Organisations (Hassan et al., 2025)

2.4 Building trust through communication

Trust is a critical component in cybersecurity. Customers must feel secure, so that their data is managed responsibly. Effective communication directly contributes to the establishment of this trust. Transparency regarding data protection policies, clear message of vulnerabilities enhances a company's credibility (Cisco, 2023).

Digital trust covers more than security, it includes perceptions of organizational ethics, transparency, and competence (Preecha, 2025). Consistent, clear, and informative communication fosters trust. Within cybersecurity awareness initiatives, trust-based communication encourages audiences to adopt safer behaviours and engage more actively with educational content (Abrahams et al 2024).

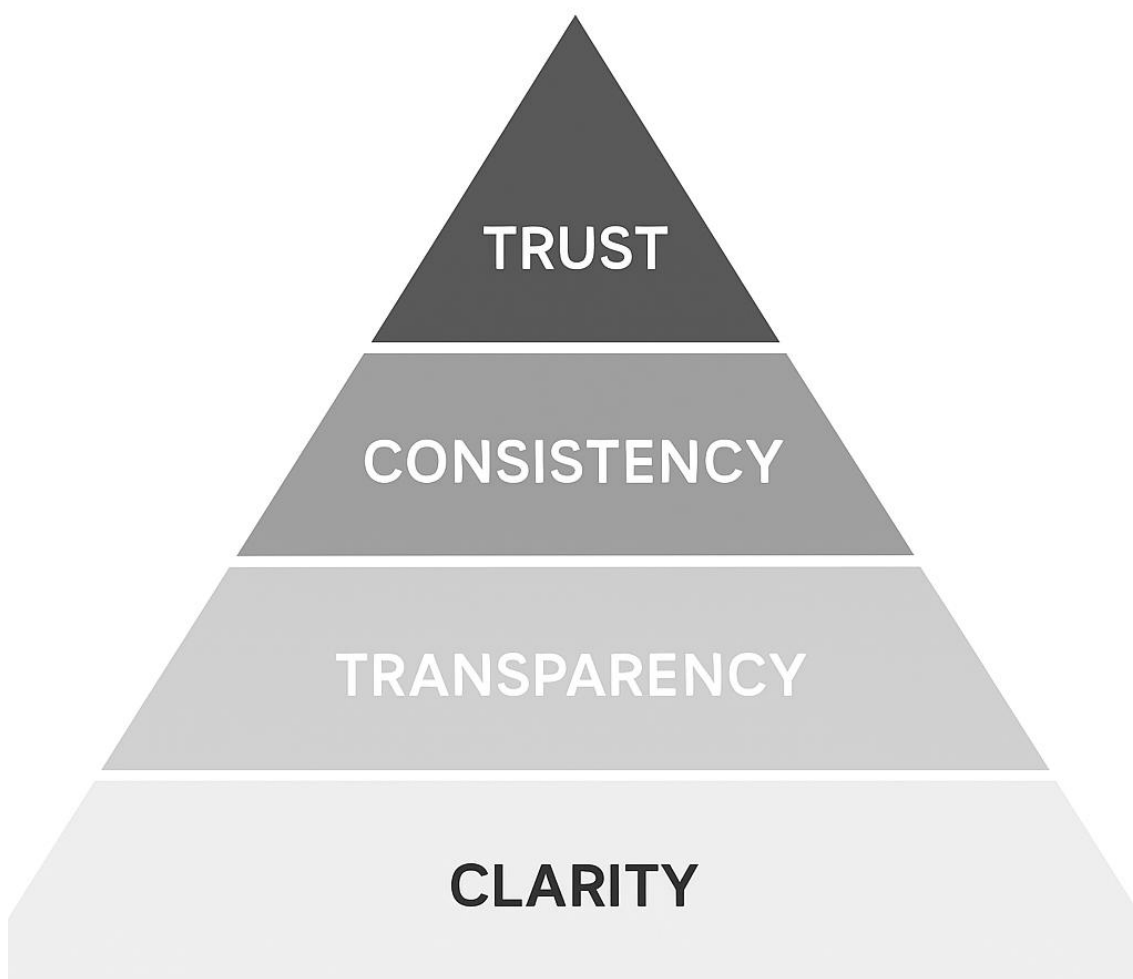


Figure 1. The digital trust pyramid

As shown in figure 1, the layered structure is formed in cybersecurity communication. On the first level lies **clarity**, referring to messages that are simple, understandable, and free of unnecessary technical jargon. Clear communication reduces cognitive friction and ensures that users can interpret security information correctly.

The second level, **transparency**, reflects how openly an organization communicates about its security practices, risks, and protective measures. Transparent communication strengthens credibility and signals honesty. An essential perk discussing sensitive cybersecurity issues.

Followed by **consistency**, emphasizing the need for repeated, stable messaging delivered across multiple channels. Consistent communication reinforces key security behaviours and reduces uncertainty. Helping users practice recommended steps over time.

Final level is **trust**, which develops when a user perceives an organization as trustworthy, competent, and communicative. Trust enables users to accept guidance, follow security recommendations. In cybersecurity awareness efforts, building trust is a growing process that depends on the strength and alignment of all underlying layers.

3 RESEARCH METHODOLOGY

3.1 Case study design

A qualitative case study design was adopted to examine the communication and awareness-building practices of *CVEHunter*. This approach was selected because, according to Yin's (2018) research, this method enables an in-depth analysis of communication strategies within a real business context and their role in enhancing cybersecurity awareness.

Empirical insights into CVEHunter's digital communication, SEO optimization, and outreach efforts were gathered through the author's direct participation during the internship.

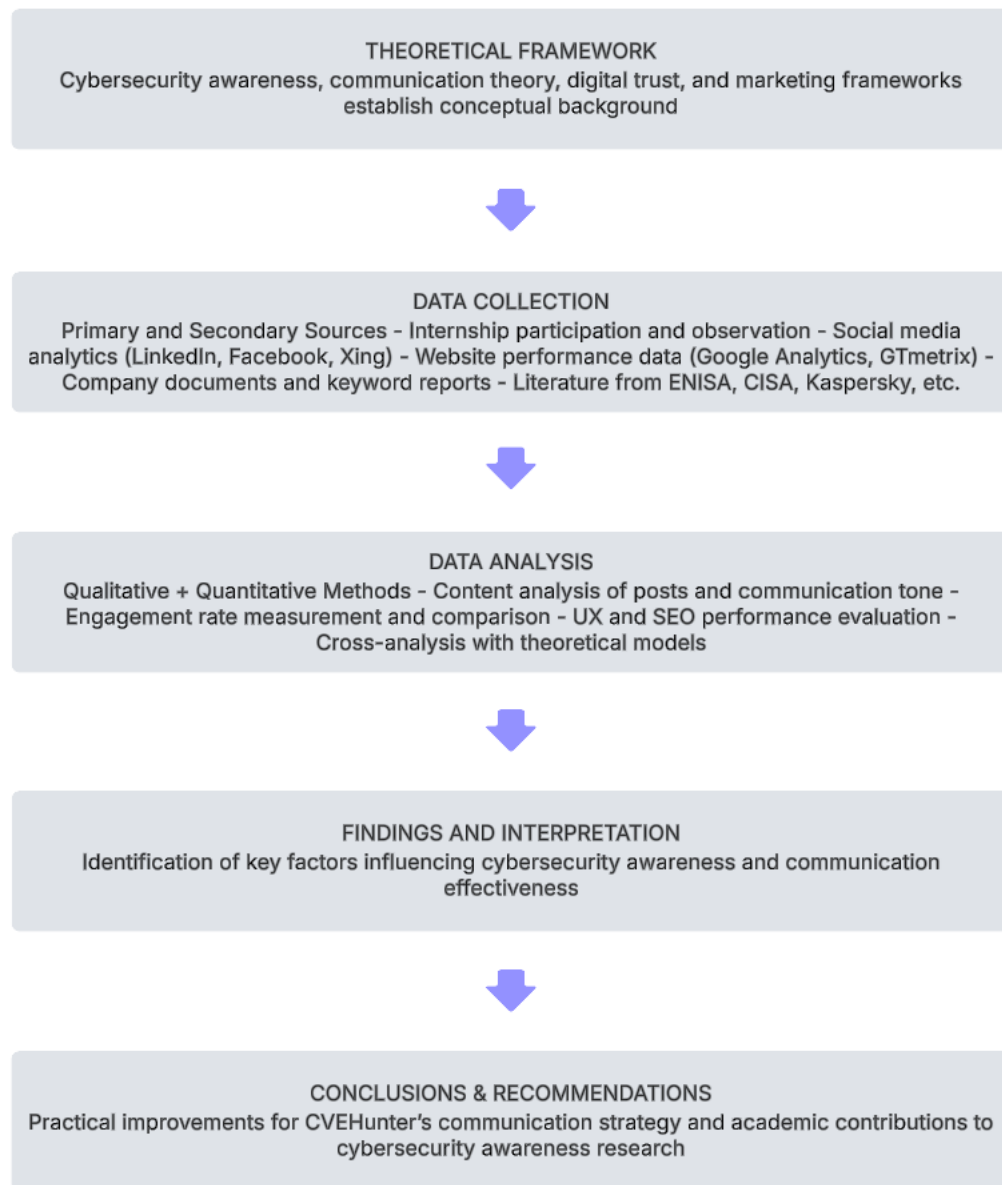


Figure 2. Research process flowchart

As shown in figure 2 the research process followed a structured sequence from theoretical framework to conclusions. This systematic flow enabled me coherence between the academic framework and the empirical analysis.

3.2 Data sources and collection methods

Data collection combined **primary and secondary sources**. Primary data was gathered through practical work activities, while secondary data supported the interpretation of communication effectiveness.

3.2.1 Primary Data

1. **Participant observation:** My internship role included creating and publishing content on LinkedIn, Facebook, Xing, and Reddit. Weekly activity logs provided continuous insights into post-performance, engagement, and audience behaviour.
2. **Social media analytics:** Metrics such as reach, impressions, and reactions were retrieved directly from platform dashboards to assess engagement effectiveness.
3. **Website analytics:** Tools like **Google Analytics**, **GTmetrix**, and **PageSpeed Insights** were used to evaluate user behavior and technical performance factors influencing bounce rate and session duration.
4. **Internal documentation:** Task records, keyword lists, and campaign notes stored in Trello were reviewed to identify planning and coordination processes.

3.2.2 Secondary Data

1. Reports from **ENISA (2024)**, **CISA (2024)**, and **Kaspersky (2023)** provided theoretical background on cybersecurity awareness communication.
2. Industry-level insights from **Edelman & LinkedIn (2024)** and **Cisco (2023)** were used to benchmark engagement and trust-building best practices.

Data Source	Type	Purpose
LinkedIn Analytics	Primary	Measure engagement and visibility
Website Audit (GTmetrix)	Primary	Evaluate performance and UX quality
ENISA Awareness Report	Secondary	Provide theoretical context
CVEHunter Reports	Primary	Understand internal marketing workflow

Table 1. Collection methods and purposes

3.3 Analytical framework

The research uses a **mixed-method analytical approach**, combining **qualitative content analysis** with **quantitative performance evaluation**.

3.3.1 Qualitative Analysis

The qualitative content analysis examined:

- Post tone, structure, and communication clarity.
- The balance between educational and promotional content.
- Audience interaction styles (comments, poll participation, shares).

Posts were categorized into key themes such as:

1. **Educational posts** – cybersecurity awareness tips or informative content.
2. **Promotional posts** – CVEHunter service introductions and product updates.
3. **Interactive content** – polls, Q&As, or discussion prompts.

This classification followed the principles of content analysis by **Krippendorff (2018)**, focusing on interpreting message meaning, consistency, and relevance.

3.3.2 Quantitative Analysis

Engagement and performance data was collected using:

- Post engagement rate (likes, comments, shares divided by impressions).
- Reach comparison across different platforms.
- Website metrics such as **bounce rate**, **average session duration**, and **page load time**.

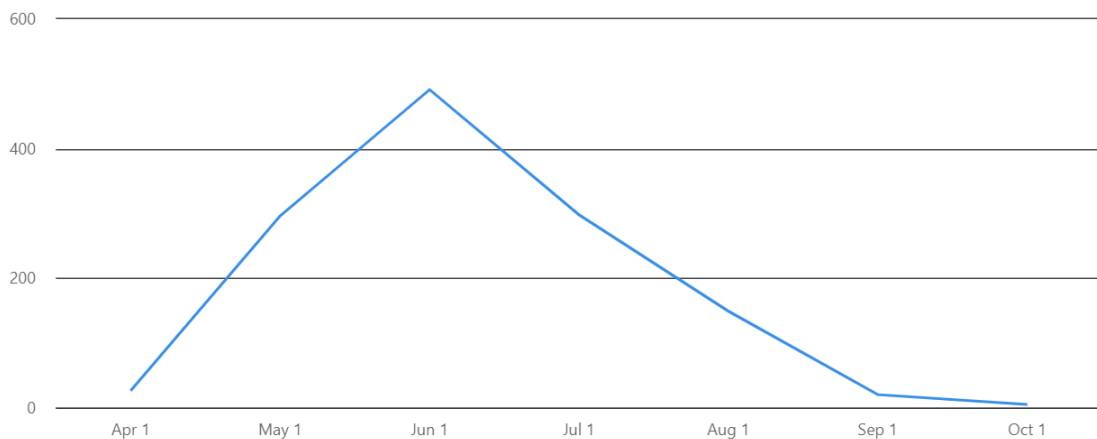


Figure 3. LinkedIn reach rate.

As shown in figure 3, using polls and informative posts, social media reach increased from May to July.

3.4 Ethical considerations

Ethical integrity was maintained throughout the research process following **Tampere University of Applied Sciences' Research Ethics Guidelines (TAMK, 2023)**.

- **Confidentiality:** Sensitive company data, internal correspondence, or client details were excluded.
- **Data usage:** All materials used for analysis were non-confidential and approved for academic use by CVEHunter.
- **AI tools:** ChatGPT, Grammarly, and Google Gemini were used responsibly for linguistic enhancement and structural organization. The author remained fully responsible for the analysis and conclusions.

4 EMPIRICAL FINDINGS

4.1 Communication and awareness practices at CVEHunter

CVEHunter's communication strategy is to build brand awareness and increase cybersecurity awareness within its audience. During my internship period (May–July 2025), the company relied on four digital channels: **LinkedIn, Facebook, Xing, and Reddit**.

The main objective was to engage potential customers, share informational content. CVEHunter's communication style reflects broader industry trends, combining educational messaging with clear promotional content to foster trust (ENISA, 2024).

CVEHunter's communication method was to approach the audience with **clear explanation of complex cybersecurity concepts**. Posts often provided simple explanations of threats such as phishing, malware, and AI-driven scams. Along with awareness tips and value propositions for the CVEHunter scanning tool.

According to CISA's (2024) recommendation to use "clear, actionable messages" when delivering cybersecurity guidance to non-technical audiences. Short-form posts (50–150 characters) were used to ensure readability and increased engagement on social-media platforms.

4.1.1 Interactive content usage

A second key element was the implementation of **interactive content**, such as polls. For example, CVEHunter published a poll titled *"What is your biggest threat when using a computer?"* with options covering scams, malware, credential theft,

and AI manipulation. This type of content is widely considered effective for increasing user involvement and gauging public concern (Edelman & LinkedIn, 2021).

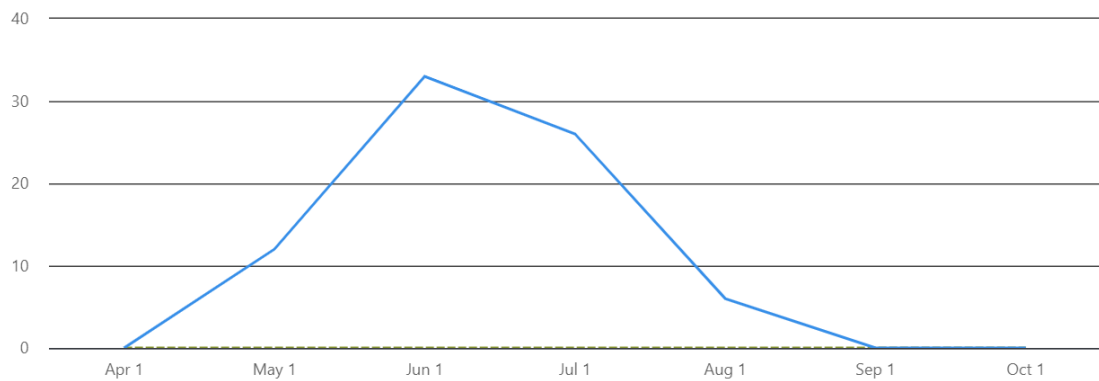


Figure 4. LinkedIn reaction rate

4.2 Author's first findings

During my internship, CVEHunter also expanded its awareness efforts by launching a **YouTube music + tech news livestream**, sharing it regularly in LinkedIn groups. It is unconventional for a cybersecurity company, main purpose was to increase brand visibility by appealing to audiences interested in ambient music and technology trends, creating an indirect but consistent point of contact. This method aligns with Campbell's (2003) findings, that consistent content exposure can improve brand recall and familiarity (Campbell, 2003).

The company's communication relied heavily on a **multi-platform posting routine**. Each week typically included:

- 1–3 LinkedIn posts
- 1–3 Facebook posts
- 1–2 Xing reposts
- occasional Reddit posts in relevant cybersecurity or IT communities

The posts ranged from educational tips to promotional updates, technical threats, and interactive questions.

According to Edelman & LinkedIn's (2024) research, LinkedIn groups are one of the most valuable B2B communication channels, so a new strategy was implemented. CVEHunter has adopted a strategy of **group-based outreach**, contacting three new LinkedIn groups each day and sharing content. While LinkedIn restricts frequent posting in groups to limit spam, the company still managed to publish 17 posts in 30 contacted groups during the period.

The consistent posting schedule is important, as SMEs often struggle with visibility due to lower algorithmic priority compared to larger brands (Statista, 2024). Therefore, an introduction of important tool has come, named Trello.

4.3 Trello as a tool

Using **Trello** for planning ensured structured oversight of content calendars, task delegation, and priority. Integrating keyword insights with post creation helped the team maintain thematic consistency across campaigns.

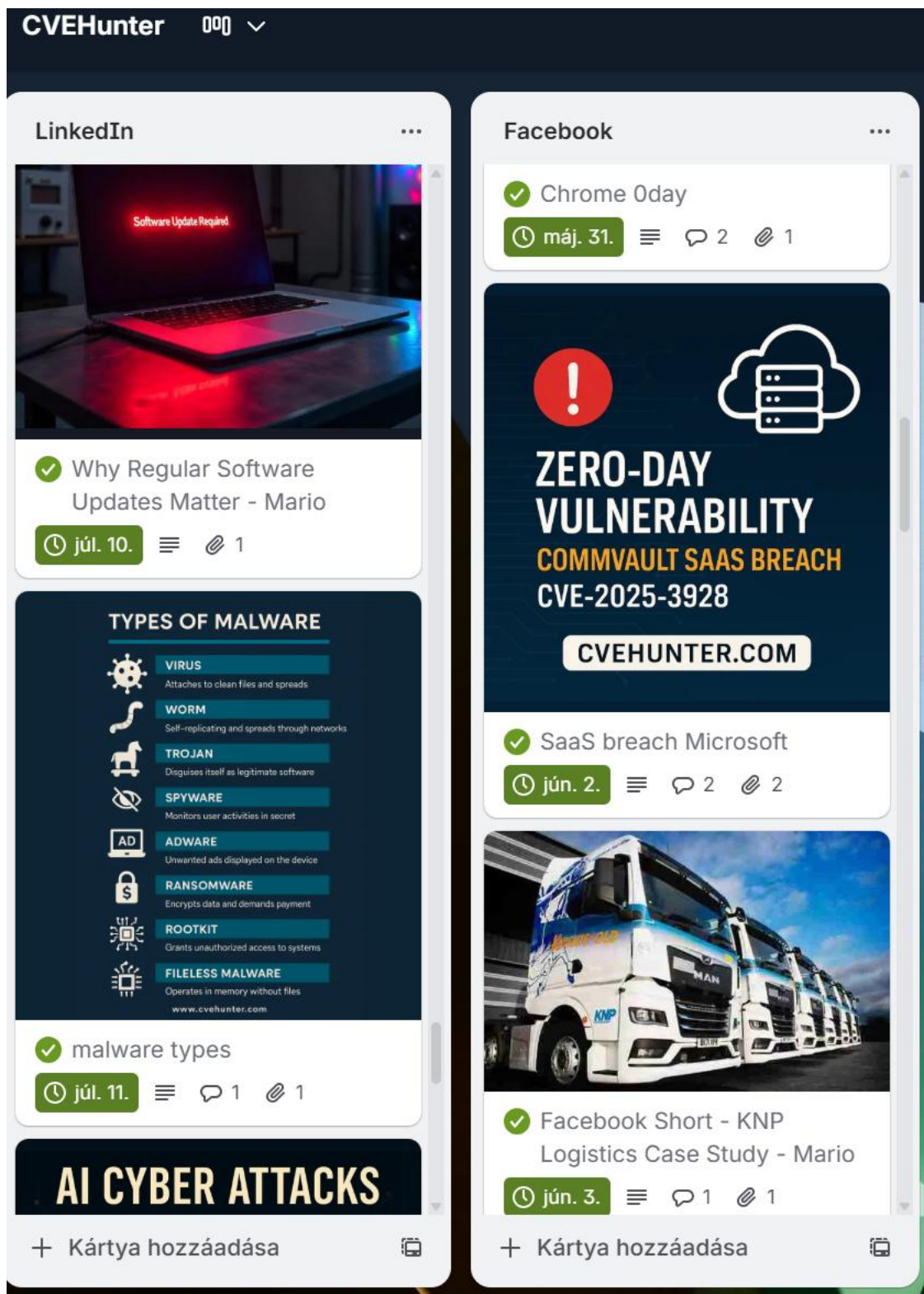


Figure 5 Trello frontview

As shown in figure 5, pre-planned posting and audience reach can make strategy planning easier that is crucial for a business.

4.4 Engagement metrics and social media performance

To assess the effectiveness of CVEHunter's cybersecurity communication, engagement metrics were collected from LinkedIn and Facebook. Although the company's audience base was initially small, measurable patterns emerged regarding content performance, platform behavior, and user interaction.

4.4.1 Overall Engagement Trends

LinkedIn was the **strongest-performing platform**, accounting for most impressions and interactions. This aligns with Statista (2024), which notes that LinkedIn remains the most relevant platform for B2B decision-makers. Posts that focused on *threat awareness* or *practical cybersecurity tips* had significantly higher engagement than purely promotional content.

For example, awareness posts generated in general 3 times more reaction than service advertisements. Polls had the highest engagement rates.

Facebook engagement was moderate but stable. Posts that contained **visual elements**, such as infographics or reels performed better than text-only messages.

Xing is the leading professional network mainly in German speaking countries. By targeting German-speaking B2B audiences, had lower engagement but proved valuable for **brand positioning**. Reposts about EU vulnerability databases (EUVD) and platform demos attracted visibility within niche cybersecurity communities.

Reddit engagement was unpredictable. Many subreddits exist with different rules and audience. Most subreddit groups have professionals, therefore for information gathering, Reddit proved to be a very liable source. While posts aligned with subreddit norms received upvotes and comments, posts flagged as promotional were removed or ignored.

- **Daily Outreach and Group Performance**

A major component of engagement-building was the daily outreach routine. The goal was to:

- contact **3 LinkedIn groups per day**,
- and **5 individual professionals or companies** for direct outreach.

LinkedIn's algorithmic restrictions and anti-spam protections meant not all posts were approved, yet CVEHunter successfully published **17 posts in 30 contacted groups**.

- **Content Type Performance**

Analysis of engagement by content type revealed predictable yet useful patterns:

- **Interactive content (polls)** — highest engagement, highest visibility.
- **Educational posts** — strong engagement, especially when referencing real threats.
- **Promotional posts** — lowest engagement unless accompanied by visuals or value explanations.
- **Tech news reports** (TechCrunch, Hacker News) — moderate engagement, useful for maintaining posting frequency.

These findings align with Edelman & LinkedIn's (2024) journal, that interactive content generates significantly higher user participation than static content.

4.5 Website analysis and UX insights

During Week 11, an analysis of the website heatmap raised important questions. *"Why users tend to leave the website after a certain period of time?"* Therefore, a comprehensive website analysis was conducted, which revealed a high bounce rate and low session duration. With the help of Google Analytics, GTmetrix and PageSpeed, website performance, UX structure, and SEO factors were evaluated.

4.5.1 Performance Metrics and Key Findings

GTmetrix results revealed a **fully loaded time exceeding 10 seconds**, significantly above the recommended threshold of 2.5 seconds according to Google's report (2023). Slow loading times are a major predictor of bounce rates, with 40–53% of users leaving a page that loads slower than three seconds. This directly impacts CVEHunter's ability to convert visitors into registered users.

Additional findings included:

- **High Total Blocking Time (TBT)** caused by JavaScript-heavy components.
- **Poor Cumulative Layout Shift (CLS)** in some sections due to image scaling.
- **Inefficient image compression**, especially for dashboard screenshots.
- Several **unused CSS and JS files**, contributing to code bloat.

Google Analytics confirmed:

- bounce rate of approximately **75–80%**,
- average session duration below **15 seconds**,
- majority of users exiting after viewing a single page.

These findings indicate that **performance issues, not the content, were the primary reason for weak website engagement**.

4.5.2 UX Structural Observations

The homepage layout had strong visual appeal, but the following UX barriers were identified:

- **The value proposition was not instantly clear**, so users could not immediately understand what CVEHunter offered without scrolling.
- **CTA buttons were not visually emphasized**, reducing conversion potential.
- **Feature descriptions were text-heavy**, which discouraged quick scanning.
- **No trust-building elements** (testimonials, security badges, partner logos) were present.

As was mentioned before, based on ENISA (2024) clear, rapid communication of security benefits is essential to building trust. CVEHunter's site partially met this requirement but needed stronger emphasis on clarity and immediate user understanding.

- **Recommended Improvements**

Based on UX and performance findings, recommendations included:

- Minimizing or deferring non-critical JavaScript.
- Implementing WebP image formats.
- Reducing above-the-fold content complexity.
- Adding **trust elements**, such as customer stories or certifications.
- Creating a clearer CTA hierarchy.

These improvements, once implemented, are expected to increase user retention and lower the bounce rate.

5 DISCUSSION

This chapter seeks to explain the author's empirical findings and reflect on their meaning in relation to the theoretical framework. To evaluate why certain communication approaches were more effective than others and how the results correspond with existing research on cybersecurity communication and awareness.

The chapter consists of three main parts. The effectiveness of CVEHunter's communication strategy is evaluated based on the empirical evidence. Second, key challenges and opportunities related to cybersecurity awareness are discussed. Finally, the findings are compared with theoretical insights to assess the quality of alignment between theory and practice.

5.1 Evaluation of communication effectiveness

According to empirical results, CVEHunter's communication practices were effective as intended on certain platforms. While several strengths were identified, particularly in awareness-oriented communication, the analysis also revealed areas for improvement.

5.1.1 Effectiveness of simple, accessible messages

The analysis showed that short, clear posts focusing on cybersecurity threats generated higher engagements. This finding suggests that users are more likely to interact with content that is valuable and easily understood by them.

This observation is consistent with earlier research highlighting the importance of clear communication in cybersecurity contexts (CISA, 2024; Parsons et al., 2017). For SME audiences, being able to understand the message without prior technical expertise appears to play a vital role in engagement. In practice, this

means that CVEHunter’s awareness-focused posts addressed not only informational needs but also users’ perceptions of cybersecurity risks.

5.1.2 Effectiveness of interactive content and polls

Polls and reels (short videos) have proved to be the most engaged content types during the analysis period. While polls had higher reactions, reels had more views. This helped users to reflect on their own security concerns. In addition, it offered CVEHunter an insight, which helped the company to understand its audience.

Due to these empirical factors, it can be assumed that cybersecurity awareness extends beyond information delivery, it involves users’ reflection on risks. Interactive content helped in a way to translate security concepts into considerations.

5.1.3 Platform-specific effectiveness

The analysis revealed that **LinkedIn** was the most effective platform for engagement, visibility, and audience relevance. This aligns with Statista (2024), which identifies LinkedIn as the primary platform for B2B communication and cybersecurity-related discussions. Posts on LinkedIn tended to reach individuals with professional interest in IT, cybersecurity, risk management, or small business development.

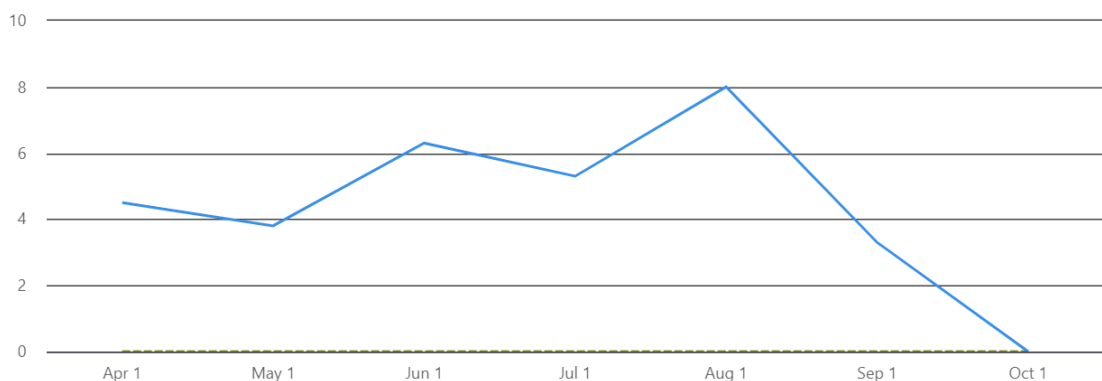


Figure 6. LinkedIn engagement rate

As shown in figure 6, the engagement rate increased in May and July due to clever targeted poll posts and topics.

Facebook had moderate engagement, particularly when posts contained visuals. This supports Cisco's (2023) findings that visual elements significantly increase message retention and security understanding.

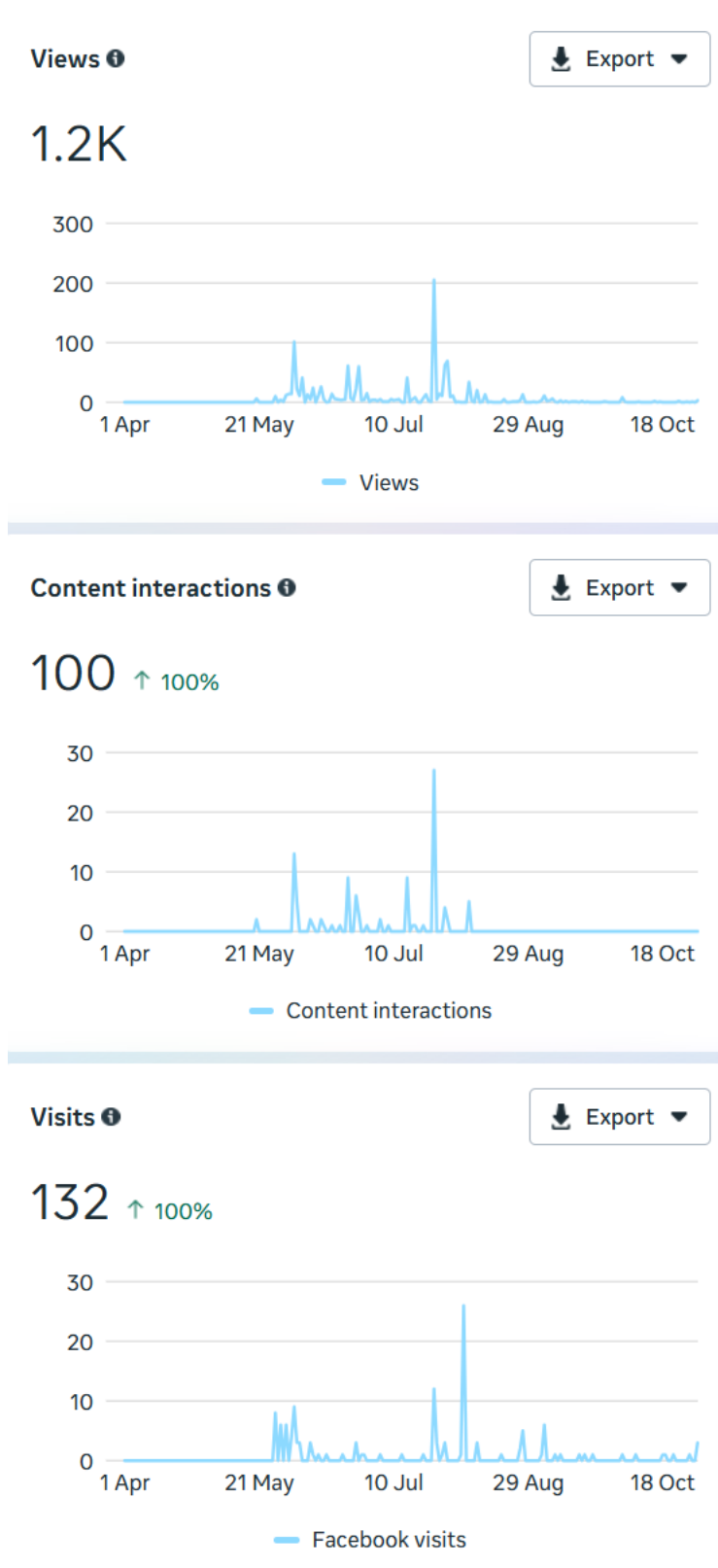


Figure 7. Facebook analytics front view

As shown in figure 8 that the same strategy can be applied to Facebook with a different modification, because LinkedIn is more formal and professional, while Facebook users prefer short daily topics and news.

Xing, while slower to respond, provided visibility in the German-speaking professional community. Reposts about EU vulnerability databases (EUVD) and cybersecurity trends increased visibility among niche audiences. This demonstrates that **platform diversification does contribute to broader awareness**, even if engagement numbers remain lower.

5.1.4 Limitations in communication effectiveness

Despite these strengths, several limitations reduced the overall impact of CVEHunter's communication efforts:

1. **Low posting frequency:** While weekly posts were created, consistent daily posting was limited by workforce and platform restrictions.
2. **LinkedIn group posting restrictions:** LinkedIn imposes strict rules to prevent spam. CVEHunter's outreach was partially constrained by moderation filters, lowering expected group-level visibility.
3. **Overreliance on short-format posts:** While effective for engagement, short posts provide limited depth and may not fully develop user understanding of cybersecurity risks or solutions.
4. **Lack of trust-building content:** Despite strong awareness communication, the content lacked elements such as client testimonials, security certifications, or case examples.

5.1.5 Overall evaluation

Overall, CVEHunter's communication was effective in raising awareness but less successful in building trust and converting engagement into long-term interaction.

5.2 Challenges and opportunities in cybersecurity awareness

Cybersecurity awareness is a complex area. It requires **changes in human behavior**, the **reduction of risk perception gaps**, and **management of cognitive biases**. The empirical findings from CVEHunter’s case point out several challenges, but also valuable opportunities for improving awareness communication.

5.2.1 Challenges identified

1. Low awareness base among users

- a. Many users demonstrated limited understanding of fundamental cybersecurity concepts. Posts that explained basic threats performed better, indicating that **awareness levels vary** across audiences.

2. Information overload and fatigue

- a. Cybersecurity can be overwhelming for non-experts.
- b. Users are frequently exposed to negative messaging (“malware,” “scams,” “data breaches”), which may create fear or avoidance. ENISA (2024) defends that excessive security messaging can lead to “alert fatigue,” reducing user responsiveness. This challenge was visible in CVEHunter’s lower performance when posting highly technical or overly repetitive messages.

3. Limited time and attention span

- a. Short posts worked well, but more complex topics—such as explaining CVEs or automated vulnerability scanning—received less engagement.
- b. This suggests that users have limited attention for complex security topics unless presented through simple narratives or visuals (Dupont et al. 2023).

4. Platform restrictions

- a. LinkedIn’s group moderation limited outreach volume. Facebook’s algorithm deprioritized business content unless boosted by visuals

or interactions. Reddit required specific formatting and community alignment to avoid removal.

- b. These constraints challenge SMEs with limited resources, reinforcing ENISA's (2024) observation that small companies often face structural barriers when promoting security awareness.

5.2.2 Opportunities identified

Despite the challenges, several strong opportunities emerged:

1. Leverage Interactivity to Improve Awareness

- a. Polls and interactive posts consistently outperformed static text. This indicates that **gamification-style communication** could be a valuable opportunity. Awareness quizzes, “spot the phishing attempt,” and interactive micro-tests could significantly increase engagement.

2. Use Visual Communication More Intentionally

- a. Visuals—such as infographics, threat diagrams, or short explainer graphics—performed better than text alone. Cisco (2023) finds that visual communication increases retention of security messages by 65%. This suggests CVEHunter could invest more in **visual cyber-security education**.

3. Build Trust Through Transparency and Human-Centric Messaging

- a. The Digital Trust Pyramid in Chapter 2 highlights clarity, transparency, and consistency. CVEHunter can strengthen trust by:
- b. showcasing real use cases, sharing anonymized scan results, highlighting certifications or affiliations, showing the team behind the platform.
- c. Such communication can help move users from awareness to trust, which research identifies as essential for behavior change (Kaspersky, 2023).

4. Multi-Platform Presence Creates Layered Awareness

- a. Even though engagement varied, CVEHunter's presence across LinkedIn, Facebook, Xing, and Reddit increased overall visibility. This multi-channel approach supports ENISA's recommendation for **distributed awareness communication**.

5. Opportunity to Address SME Pain Points

- a. SMEs struggle with cybersecurity due to lack of staff, lack of time, lack of training, fear of complexity.
- b. CVEHunter can position itself as a **simple, affordable, one-click solution**, making awareness content linked to their value proposition.

5.3 Comparison with theoretical Insight

5.3.1 Alignment with theoretical framework

1. Awareness as a human-factor issue

- a. The empirical findings support the theory that cybersecurity awareness is largely driven by **human factors**, including clarity of information, risk perception, emotional response, relevance of content.
- b. Parsons et al. (2017) and ENISA (2021, 2024) emphasize that awareness must be contextualized and communicated clearly. CVEHunter's experience confirmed this: simple threat explanations worked best.

2. Communication theory and message clarity

- a. The effectiveness of simple messages aligns with Shannon–Weaver's communication model, particularly the need to reduce "noise" (technical jargon, abstraction). CVEHunter's shorter posts reduced noise and increased comprehension.

3. Trust-building theory

- a. Chapter 2's Digital Trust Pyramid identified clarity, transparency, and consistency as foundations of digital trust. Empirical findings confirm:
- b. clarity was strong (high-performing short posts)
- c. transparency was moderate (few posts explained how the scanning tool works)

- d. consistency was limited (posting frequency varied)
- e. This confirms that **trust-building is partially achieved but not yet complete**.

4. Multi-channel communication

- a. ENISA (2024) recommends multi-channel messaging to reach wider audiences. CVEHunter successfully applied this strategy across four platforms. The empirical findings confirm the theory: different channels reach different segments, and diversification increases visibility.

5.3.2 Areas of divergence from theory

1. Resource limitations

While theory suggests high posting frequency, SMEs often lack resources to maintain it. CVEHunter's case demonstrates that resource constraints limit consistency.

2. Platform restrictions (Real-World Noise)

Theoretical communication models rarely account for social media moderation rules. LinkedIn's restrictions acted as "external noise," limiting message distribution.

3. Behavior change requires long-term effort

Theoretical models emphasize behavior change as a long-term process. The internship timeframe—two months—was too short to observe significant changes in user behavior beyond engagement metrics.

4. Awareness does not guarantee conversion

While theory suggests awareness improves security outcomes, the empirical findings showed **awareness does not automatically translate into subscriptions**, supporting Kaspersky's (2023) claim that users often remain passive despite increased understanding.

5.3.3 Overall comparison

The overall alignment between theory and findings is strong:

- Theoretical principles correctly predicted which types of communication are effective.
- Real-world constraints explain why some theoretical recommendations are difficult to implement for SMEs.

The comparison confirms that cybersecurity communication must remain **adaptable, user-centered, and resource-conscious**.

6 RECOMMENDATIONS

This chapter provides practical, research-based recommendations for CVEHunter to strengthen its cybersecurity communication, improve user engagement, and support long-term awareness-building among professional and business audiences. The recommendations are derived from empirical findings and discussions, and are designed to be realistic and implementable for a small or medium-sized cybersecurity company.

6.1 Implement a structured, multi-level communication strategy

CVEHunter's communication activities would benefit from a more clearly structured, multi-level approach that aligns content with different stages of the awareness and decision-making journey. Based on the findings, communication can be organized around three levels:

6.1.1 Awareness-level communication

At this level, the focus is on helping users recognize cybersecurity threats and understand their relevance. CVEHunter should continue producing:

- short, simple, threat-focused posts
- polls and interactive questions
- brief news summaries with clear takeaways

The goal is to increase visibility and ensure that cybersecurity feels relevant and understandable rather than abstract or overly technical (CISA, 2024; ENISA, 2024).

6.1.2 Trust-building communication

Once awareness is generated, communication should deepen understanding and build confidence in CVEHunter as a reliable partner. This level should emphasize:

- transparency about how the scanning process works
- simple diagrams or dashboard examples that show what users receive
- anonymized case examples or testimonials that highlight real benefits

This responds directly to the current gap in transparent, trust-enhancing content and supports the trust-building dimensions discussed in the Digital Trust Pyramid model.

6.1.3 Conversion-level communication

Finally, communication should encourage users to take action, such as signing up for a trial or subscribing. For this level, CVEHunter should clearly present:

- calls-to-action such as “Scan your website now”
- concise explanations of pricing, target users, and use cases
- short demo videos or screenshots illustrating the process from sign-up to report delivery

A structured multi-level strategy helps ensure that communication does not jump directly from awareness to sales but instead guides users through a logical path: understanding, trusting, and then deciding to act.

6.2 Increase the use of visual and interactive content

According to empirical findings, visual and interactive content significantly outperforms text-only messages. Therefore, CVEHunter should:

- publish **weekly infographics** that explain common threats, such as phishing in simple visual form
- convert key findings from the website audit into clear charts or diagrams that show, for example, how performance improvements reduce bounce rates
- use short “**Did you know?**” graphics with one key statistic or insight per post
- create **visual case examples**, showing before-and-after states for vulnerabilities discovered and resolved
-

Previous studies show that visual information is processed faster than text alone, particularly in complex cybersecurity contexts (Cisco Systems, 2023). Such information is also retained more effectively. For small companies, even simple, template-based visuals can therefore have a substantial impact.

6.3 Develop a consistent posting routine across platforms

The analysis showed that posts were generally well designed. However, posting frequency remained irregular. This inconsistency undermines both algorithmic performance and user expectations. To improve consistency, CVEHunter should aim for a realistic and stable posting schedule, such as the following.

- 3 posts per week on LinkedIn
- 2 posts per week on Facebook
- 1 post per week on Xing
- targeted engagement on Reddit once or twice per month in suitable communities

Scheduling and cross-posting are supported by Trello and Facebook business page site. Consistent communication helps reinforce core security messages

over time. It also contributes to building digital trust, as users perceive the company as active, reliable, and present in relevant channels.

6.4 Improve transparency and humanize the brand

Trust-building emerged as a relative weakness in the current communication. This was observed despite the presence of strong awareness-level content. To address this gap, CVEHunter should increase transparency and human elements in its messaging in the following ways.

- introducing team members and their roles in short posts or “meet the team” formats
- sharing “behind the scenes” glimpses of how scans are run, tested, or improved
- explaining the scanning pipeline in simple, step-by-step diagrams
- emphasizing how user data is handled securely and responsibly
- sharing anonymized success stories or typical scenarios, such as “A small agency scanned its website and found X critical vulnerabilities”

These actions should help users see CVEHunter not as a black-box tool, but as a trustworthy partner staffed by real professionals.

6.5 Address website UX and performance issues as a priority

Regardless of the quality of communication on social channels, a slow or confusing website can undermine both awareness and trust. The website analysis identified slow loading times, layout shifts, and unclear value messaging on the landing page. To address these issues, several priority actions should be taken.

- optimizing images and scripts to significantly reduce loading time
- improving layout stability to avoid distracting visual shifts
- restructuring the homepage so that the main value proposition (“what CVEHunter does and for whom”) is visible immediately without scrolling
- placing clear CTA buttons (e.g. “Start Scan,” “Get Demo”) in prominent positions

- adding trust indicators, such as security badges, expert quotes, or client testimonials

Avoiding miscommunication, clarifying the user journey can help CVEHunter to create a more effective conversion funnel. Guiding great share of incoming traffic toward trial usage or subscription. Usability and performance are integral components of effective security communication (ENISA, 2024)

6.6 Create a cybersecurity knowledge hub

To position itself as a long-term partner rather than a one-off scanning service, CVEHunter could benefit from developing a small and focused online knowledge hub. Such a hub would support ongoing engagement and awareness-building. It could include the following elements:

- short cybersecurity tutorials tailored for SMEs
- explanations of common vulnerabilities in non-technical language
- practical best-practice guides (e.g. checklists for secure website maintenance)
- downloadable resources such as PDF checklists or “first steps after discovering vulnerabilities”

Such a hub would support ongoing awareness-building, attract organic traffic through SEO, and reinforce CVEHunter’s role as an educator and trusted advisor in addition to being a tool provider.

7 CONCLUSION

The purpose of this thesis is to explore and gather information, that how cybersecurity awareness can be communicated to daily users and business entities. The author's internship at CVEHunter served the purpose of empirical data gathering, practical experience from social media analytics and website performance tools. With the combination of theoretical knowledge, provided by other experts on the topic.

Based on the findings, it can be assumed that cybersecurity communication for SMEs not just primarily a matter of technical jargon, but rather clarity of the topic. The knowledge that the analysed platform provided, it was concluded that users engaged in contents that were short and easy to understand. Implementing interactive elements has increased the engagement rate, this finding proved to be particularly valuable. Both CVEHunter and the audience had an insight into other user's vulnerabilities they feared the most, this helped to transform the contents from passive information gathering to active awareness spreading.

Findings proved that awareness alone is not enough. While CVEHunter's communication strategy succeeded in raising visibility and generating engagement, especially on LinkedIn, it was less effective in converting this interest into deeper trust and concrete action. This underlines the strong coherence between communication, trust, and usability in digital cybersecurity services.

From a methodological perspective, the case of CVEHunter approach proved suitable for understanding and analyzing both quantitative and qualitative dimensions of cybersecurity communication in a real business environment.

Social media metrics and website analytics were gathered by the from the internship. However, the study is also limited by its relatively short time frame, dependence on platform-level analytics, and focus on a single company. Cybersecurity awareness and behavior change are long-term processes, therefore only assumption can be made based on relevant research.

The author's main suggestion is to extend the analysis to a longer observation period. That should include comparative cases from other companies. That could help other business entities to understand how target audience react and consume cybersecurity messages.

The thesis offers both practical and academic value despite previously mentioned limitations. In the case of CVEHunter, many recommendations are based on theoretical research. However, it is explained in a concrete manner that highlights the importance of communication skills and behavioural insights.

Based on the author's empirical and theoretical findings, this thesis proves that improving cybersecurity awareness means more than identifying technical threats. It also requires connecting security experts with daily users. By reaching an accessible language, a company like CVEHunter can play a key role in the field of cybersecurity.

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